

Harshawardhan Shrivastava

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SUMMARY

AI-native builder focused on shipping real AI systems end to end: real-time voice agents, hybrid RAG pipelines, NLP-driven retrieval, MCP servers, and automated evaluation harnesses. Experienced in debugging asynchronous systems, benchmarking retrieval and latency, and turning prototype failure modes into measurable engineering improvements. 3x national hackathon winner with a bias toward building and iterating on real problems under time constraints.

PROJECTS

- miniVoxSetu – Real-Time Multimodal Voice AI Agent for Banking**

2026
- *Python, FastAPI, WebSockets, Groq, Deepgram, Qdrant, PyTorch* | [GitHub](#) | [Live Demo](#)
 - Engineered a real-time multimodal voice AI agent for Indian banking, combining streaming STT/TTS, NLP-based semantic analysis, acoustic emotion detection, hybrid RAG, and asynchronous processing; reduced end-to-end conversation latency from 1,250ms to approximately 390ms.
 - Built and evaluated a hybrid RAG engine combining BM25 and vector search with Reciprocal Rank Fusion; automated benchmarks improved Hit@1 from 68.4% to 93.8% and reached 100% Hit@3.
 - Designed a 7-layer barge-in system using AudioWorklet VAD, client queue flushing, WebSocket interruption signals, generation-ID gating, and stale-task pruning; reduced interrupt-to-silence latency to under 280ms and resolved an asynchronous audio race condition.
- GitHub Analyzer – MCP Server & Agentic Code Review Tool**

2026
- *Python, MCP SDK v2, Gemini 2.5 Flash, Groq LLaMA 3, GitHub REST API, Docker, Azure* | [GitHub](#) | [Live Demo](#)
 - Built a Model Context Protocol server from scratch using the official MCP Python SDK v2, exposing a typed `analyze_repo` tool over JSON-RPC/stdio for Claude Desktop, Cursor, Kiro, and MCP Inspector.
 - Designed a two-model analysis pipeline using Gemini 2.5 Flash for deep repository analysis and Groq LLaMA 3 8B for fast structured scoring, evaluating repositories across 8 weighted engineering categories.
 - Added Streamable HTTP transport and containerized the service with Docker, deploying to an Azure VM for remote hosting; extended the same MCP tool from local subprocess execution to a deployable cloud service.
- HireReady – Multi-Modal AI Interview & Placement Platform**

2026
- *Team Project – FastAPI, React, Firebase, ChromaDB, Groq, Gemini, Whisper, MediaPipe* | [GitHub](#) | [Live Demo](#)
 - Built an end-to-end AI placement platform combining resume ATS analysis, GitHub codebase verification, semantic job matching, and multimodal AI mock interviews into a composite PlaceScore.
 - Designed a deterministic and neural ATS engine combining keyword matching, semantic similarity, format analysis, skills coverage, experience relevance, and education matching into a reproducible 0–100 score.
 - Implemented ChromaDB-based semantic job matching, Whisper transcription with hallucination filtering, and adaptive interview generation grounded in resume and GitHub context.

TECHNICAL SKILLS

Languages: Python, SQL
Generative AI: LLMs, NLP, Agentic Pipelines, RAG, Hybrid Search, MCP, Prompt Engineering, Evaluation Harnesses
Backend: FastAPI, Firebase, WebRTC, WebSockets
Databases: Qdrant, ChromaDB, Firebase, MySQL
Cloud & DevOps: Azure, Docker, Git, GitHub
Tools & Practices: Prompt Engineering, Responsible AI, n8n
ML Models: DistilHuBERT, Librosa, Ollama

ACHIEVEMENTS

3x National Hackathon Winner – NASA Space Apps Challenge, Qubitx 2025, KV Sangathan
Finalist in 8 national hackathons, including Smart India Hackathon 2025
President, Student Activity Council, OIST Bhopal (2025 – Present)

EXPERIENCE

- Freelance AI Automation**

Dec 2025 – Present
- Independent*

Remote

 - Delivered 5+ AI-powered workflow automations for small businesses, including client pipelines, lead capture, and document processing.
- Technology Intern**

Nov 2025 – Jan 2026
- HaronIndia Pvt. Ltd.*

Remote

 - Implemented AI features for farmers alongside the founding team.

EDUCATION

Oriental Institute of Science and Technology

Bhopal, India

Bachelor of Technology in Computer Science & Engineering; CGPA: 8.02

2023 – 2027